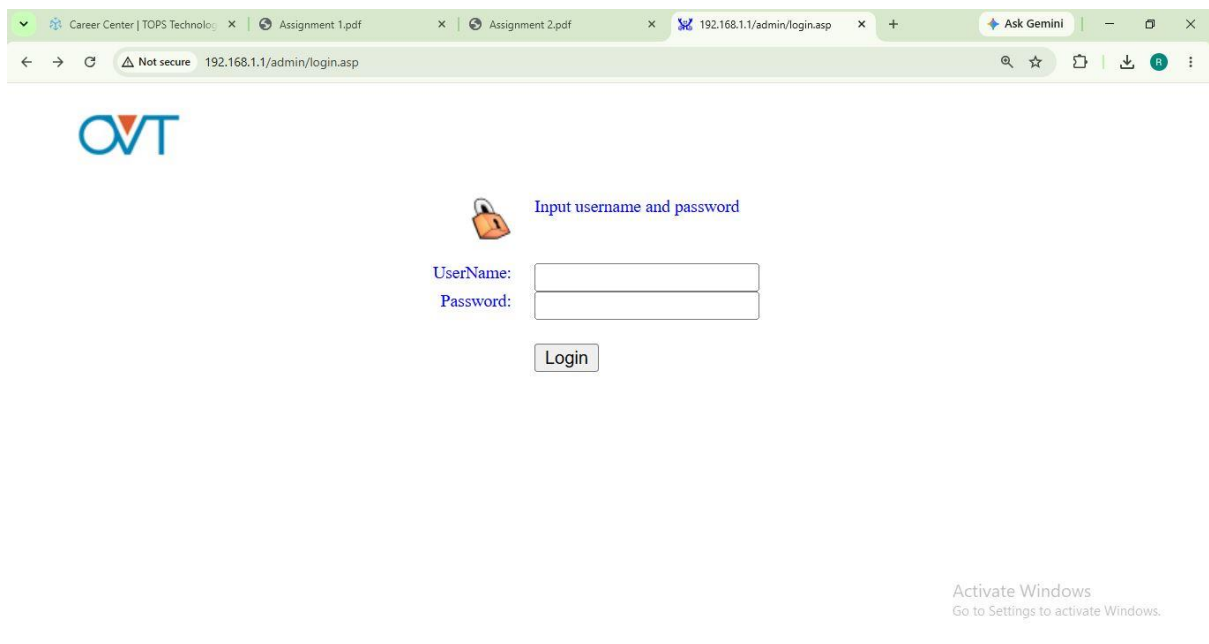


1. Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page.



2. Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

My Laptop configurations:

- **IP address :** 198.168.1.7
 - **Subnet mask:** 255.255.255.0
 - **Default gateway:** 192.168.1.1
3. Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access.

```
Command Prompt

Link-local IPv6 Address . . . . . : fe80::553e:50fb:ad76:95c%7
IPv4 Address. . . . . : 192.168.31.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::71c2:c55f:57c7:7cfe%20
IPv4 Address. . . . . : 192.168.231.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :

Wireless LAN adapter Wi-Fi:

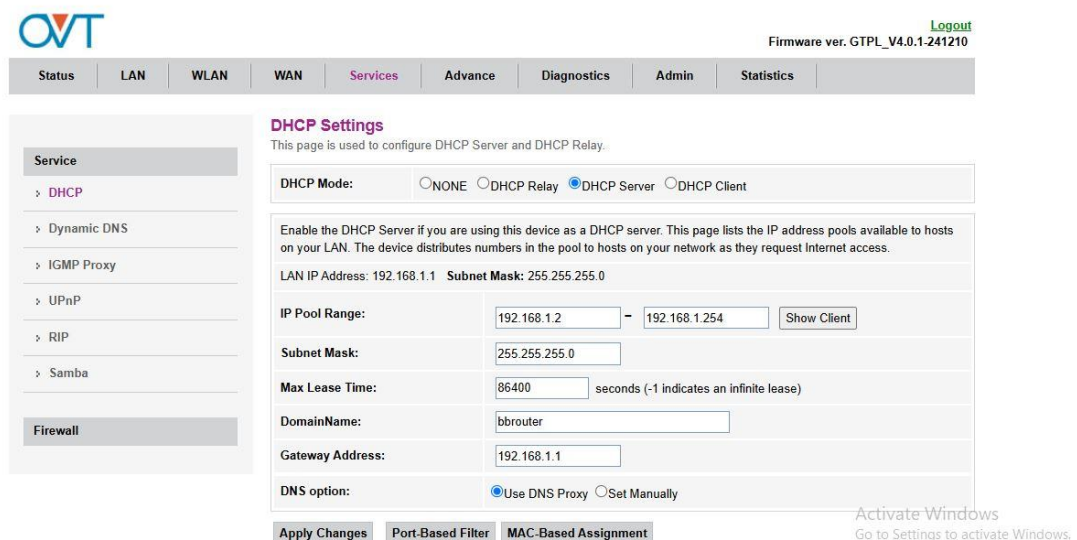
Connection-specific DNS Suffix . :
IPv6 Address. . . . . : 2402:a00:409:815:470a:b7b5:7d21:bdc6
Temporary IPv6 Address. . . . . : 2402:a00:409:815:9c5a:dbfa:c3ac:d627
Link-local IPv6 Address . . . . . : fe80::bcff:c2db:3220:39b7%6
IPv4 Address. . . . . : 192.168.1.23
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::6231:92ff:fe16:8c45%6
                             192.168.1.1

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

:\Users\rohan>
```

4. In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.



- If two devices are assigned the same IP address within this range, an IP address conflict occurs. This can cause one or both devices to lose network connectivity, experience intermittent connection problems, or be unable to communicate properly with other devices on the network.